21-242 Recitation 12

Rank–nullity — Q4. T(U) U, and equality

Course-level solution, with a later-course alternative when useful

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Prompt

Let $T : V \to W$ and $U \subseteq V$ a subspace. Show that $\dim(T(U)) \leq \dim(U)$. When does equality hold?

Course-level solution

The restriction $T|_U : U \to T(U)$ is linear and surjective, with kernel $U \cap \ker T$. Rank-nullity yields

$$\dim T(U) = \dim U - \dim(U \cap \ker T) \leq \dim U.$$

Equality holds iff $\dim(U \cap \ker T) = 0$ iff $U \cap \ker T = \{\vec{0}\}$ iff $T|_U$ is injective.
Special cases: equality for every U iff $\ker T = \{\vec{0}\}$ iff T is injective. Equality for $U = V$ is the statement $\dim \operatorname{im} T = \dim V$, again injectivity of T.